



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · 4 categories · Strategic insights from data

Dataset at a Glance

3,900

Transactions

18

Features

50

Locations

37

Missing Values

Review Rating only

Python Data Preparation

01

Load & Explore

pandas: `df.info()`, `.describe()`

02

Clean

Median imputation for Review Rating

03

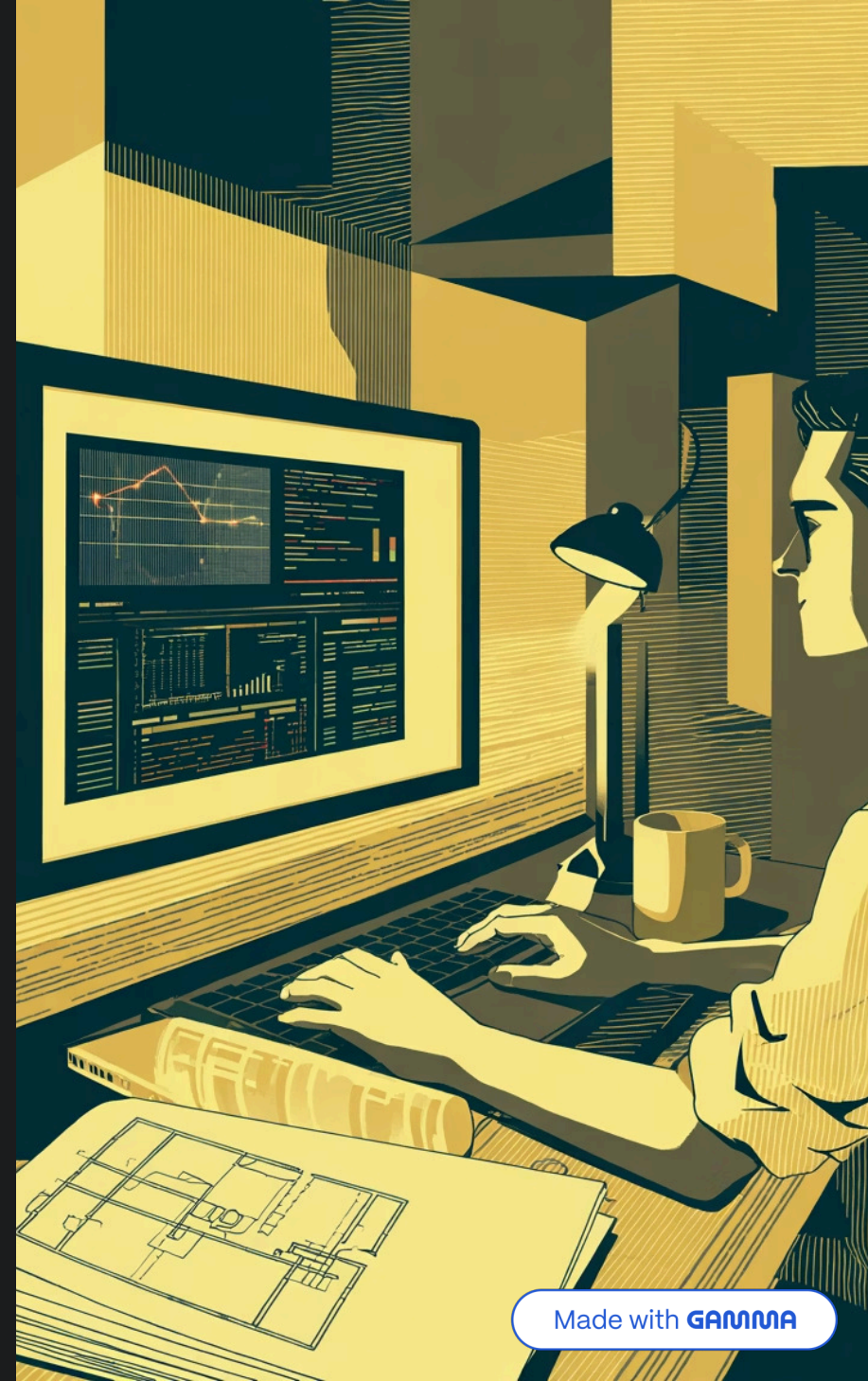
Engineer

age_group bins, purchase_frequency_days

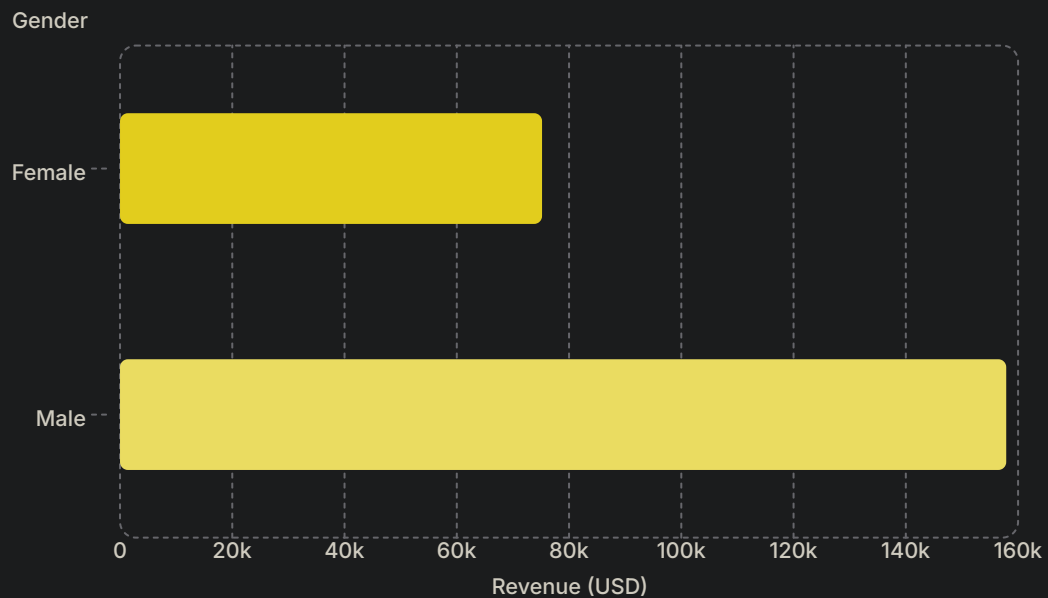
04

Load

PostgreSQL integration



Revenue by Gender



Key Insight

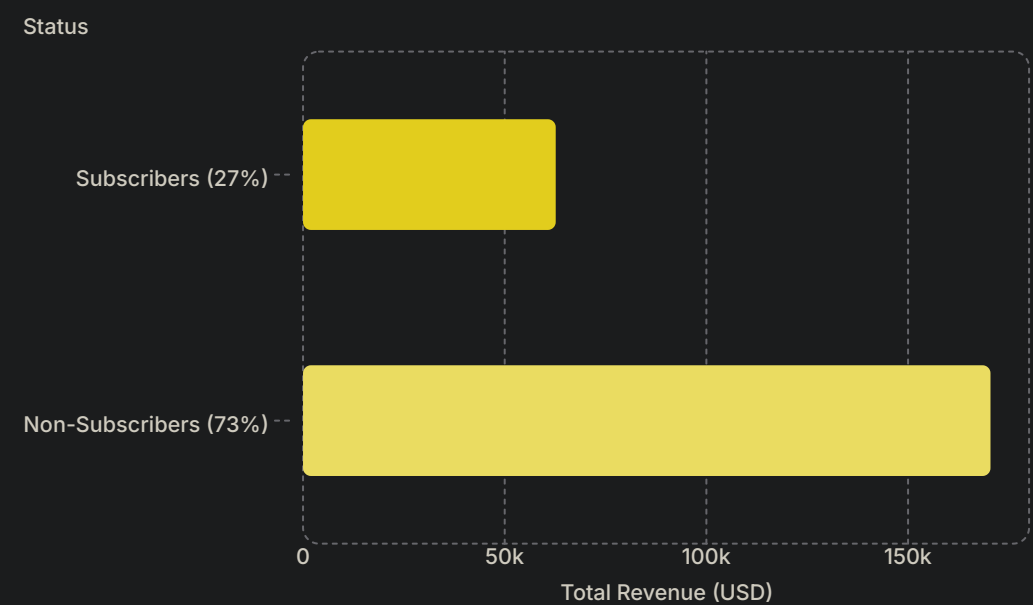
Male customers generate **2.1× more revenue** than female customers.

Shipping & Subscriptions

Shipping Type — Avg Spend

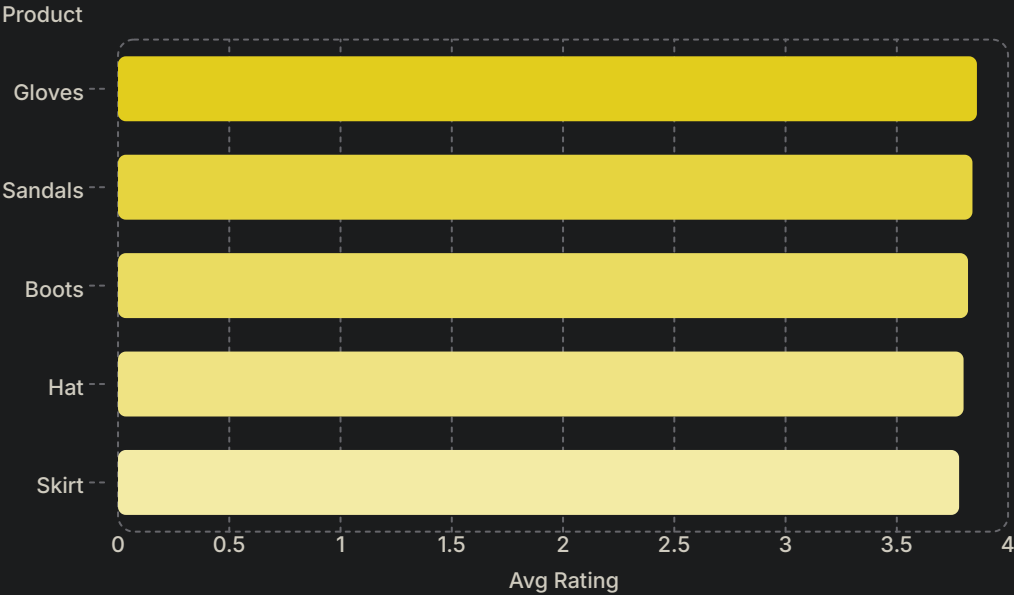


Subscription Revenue

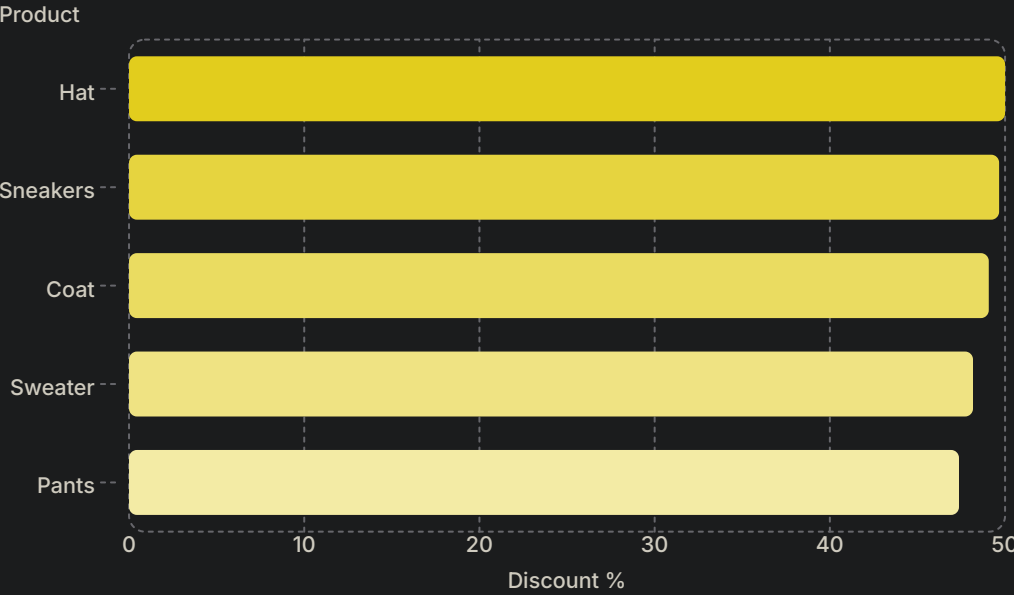


Top-Rated & Discount-Dependent Products

★ Top 5 by Rating



🏷️ Highest Discount Rate



Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

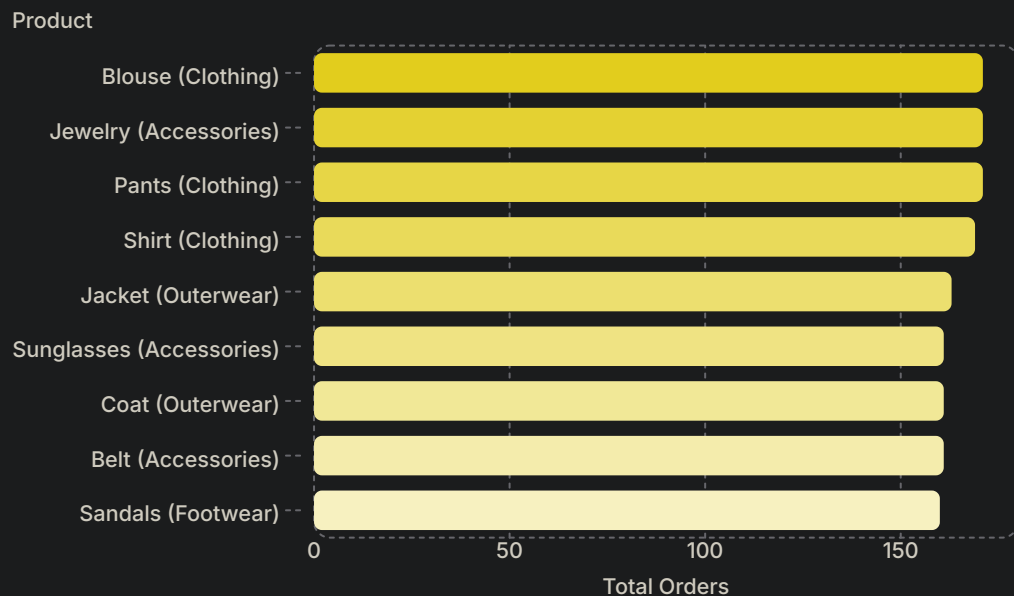
New

83 customers

839 high-spenders used discounts yet spent **above average**.



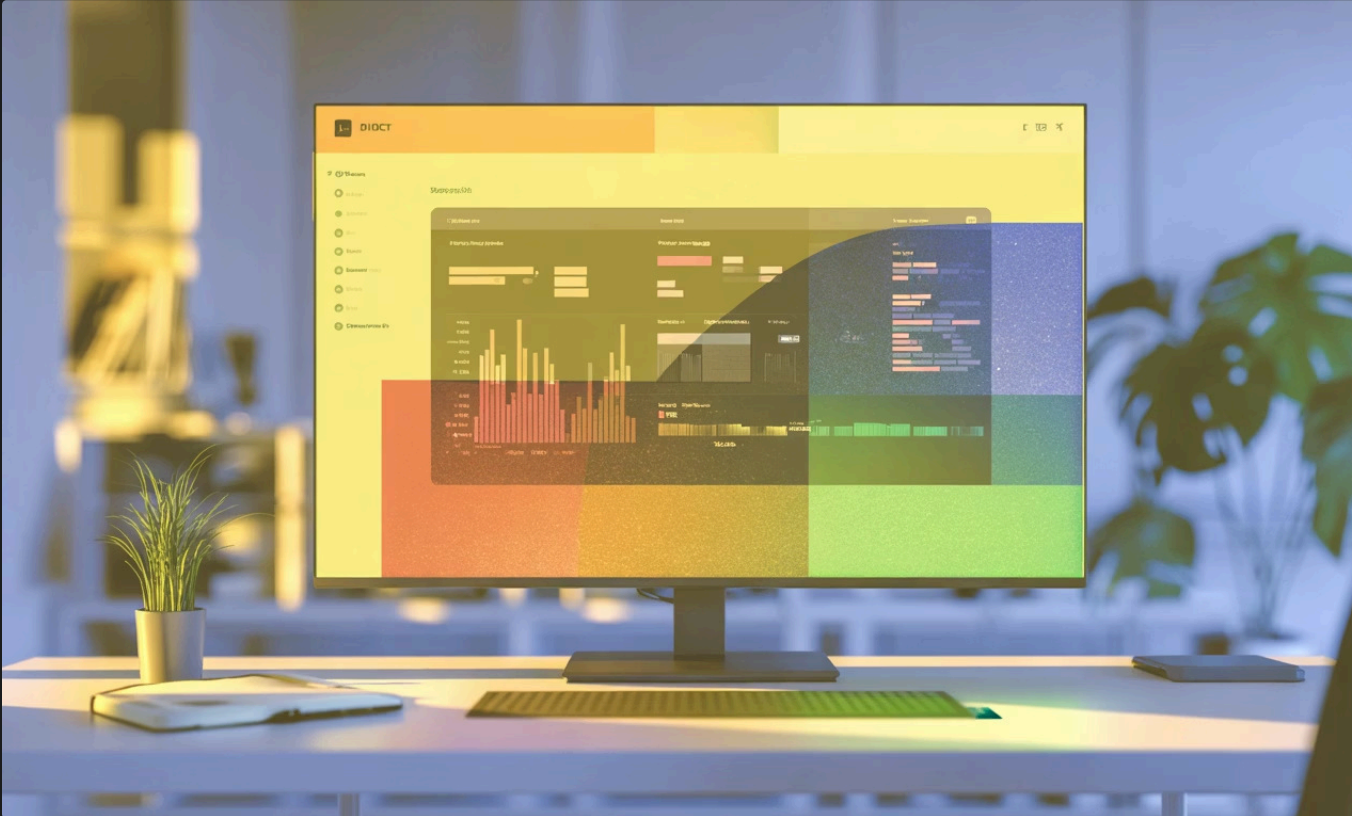
Top Products by Category



Category Leaders

- **Clothing:** Blouse, Pants, Shirt
- **Accessories:** Jewelry, Sunglasses, Belt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat

Power BI Dashboard



Key KPIs

- 3,900 customers
- \$59.76 avg purchase
- 3.75 avg review rating

Filters

Subscription · Gender · Category ·
Shipping

Visuals

- Revenue by Category & Age Group
- Subscription split: 27% Yes / 73% No

Business Recommendations



Boost Subscriptions

Exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers



Review Discount Policy

Balance sales with margins



Targeted Marketing

Focus on high-revenue age groups & express shippers

